Exploring the Construct Validity of Letters of Recommendations in Graduate Admissions: The Hidden "Unknowns" Within

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What do "unknown" ratings in letters of recommendation mean?

- Suggests lower qualified applicant
- Indicates shorter relationship with recommender
- Given to infrequently observed skills

Introduction

- Letters of recommendation (LORs) are ubiquitously used in academia as they are assumed to predict performance. Despite their wide use, LORs are controversial in their reliability and validity.
- Kuncel, Kochevar, and Ones (2014) found that although many correlations between LORs and performance were small, there were "reasons for hope" that LORs can be powerful tools.
- In a business setting, Hanisch (1992) found "unknown" responses to have negative nuances.



- This study examined the construct validity of "unknown" ratings on LORs.

Research Question

What is the relationship between the number of "unknowns" given by a recommender and other applicant information?

Hypotheses

Hypothesis 1: More "unknowns" will tend to be associated with weaker applicant credentials.

Hypothesis 2: More "unknowns" will be given by recommenders who had a shorter relationship.

Hypothesis 3: Characteristics dealing with unobservable behaviors in a lab setting will receive more "unknowns."

Methods

- Subjects were 194 applicants from a graduate biomedical engineering program at a large research university.
- Archived applications were deidentified so researchers could extract information, such as undergraduate GPA (UGPA) and GRE scores.
- Applicants' research accomplishments were quantified on an anchored scale. "Unknown" responses were totaled across 17 qualities.

Results

- "Unknown" responses significantly correlated with research accomplishments, GRE-Analytical and Verbal scores, and length of recommender-applicant relationship.
- International and domestic applicants differed in GRE-Analytical and Verbal scores and ratings.
- Qualities 'professional experience,' 'graphic expression,' 'leadership ability,' and 'teaching potential' received the most "unknowns."[a]

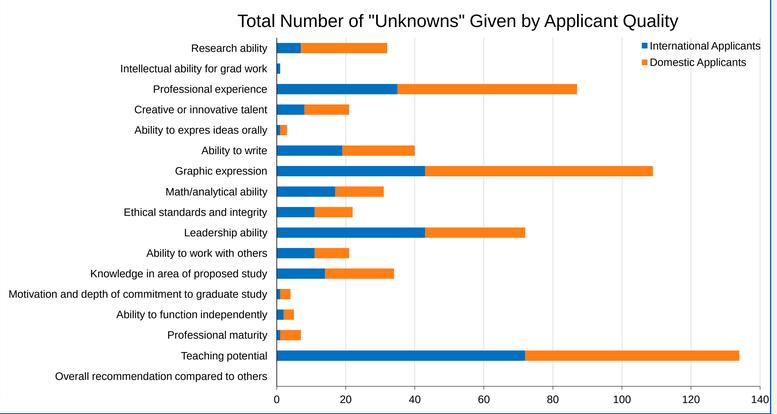
Table 1

Means, standard deviations, and correlations of information from all applicants.

Variable	*Mean*	*SD*	1	2	3	4	5	6	7
1. UGPA-C									
2. GRE-V	154	6.76	.13						
3. GRE-Q	165	5.51	.19**	.29***					
4. GRE-A	3.73	0.71	.11	.46***	-.08				
5. Unknowns	3.21	3.62	-.11	.14*	-.12	.21**			
6. Rating	4.57	0.82	.20**	.00	.07	.05	-.06		
7. Years Known	2.72	1.90	.00	-.20*	-.10	-.13	-.16*	.21***	
8. Research Accomplishment	4.41	1.13	-.10	.00	.08	-.17*	-.24**	.07	.14

Note. $N = 194$. UGPA-C was calculated in z-scores due to varying numeric grading scales. * $p < .05$. ** $p < .01$. *** $p < .001$.

Results (cont.)



Discussion

- "Unknown" responses within LORs are not neutral. They reflect an applicant's qualification and the relationship with a recommender.
- The next steps of our research entail removing the less relevant qualities[a] and reexamining the relationship between "unknowns" and applicant information.
- Future researchers should seek to understand why recommenders respond with "unknowns."
- Limitations include having access to information about a single graduate program at one university. As a result, other programs should attempt to replicate our findings.

References

Hanisch, K. A. (1992). The Job Descriptive Index revisited: Questions about the question mark. *Journal of Applied Psychology, 77*, 377-382. doi:10.1037/0021-9010.77.3.377

Kuncel, N. R., Kochevar, R. J. and Ones, D. S. (2014), A meta-analysis of letters of recommendation in college and graduate admissions: Reasons for hope. *International Journal of Selection and Assessment, 22*, 101–107. doi: 10.1111/ijsa.12060

[Untitled photograph of question marks]. Retrieved from http://blogs.babycenter.com/mom_stories/9-things-you-hope-your-kids-never-ask-you/